

Modeling Practice: Regression & Regularization in R

Objective

Predict the number of college applications received using institutional characteristics from the ISLR College dataset. The response was log-transformed, data were split 80/20 into training and test sets, and candidate models were evaluated on the held-out test data.

Models compared

Ordinary least squares, ridge regression, and elastic net. The workflow used preprocessing and cross-validation to tune regularized models.

Reported held-out results

Model	RMSE	R-squared	MAE
Ridge Regression	0.2651	0.9534	0.2075
Linear Regression	0.3166	0.9206	0.2532
Elastic Net	0.6720	0.9480	0.5492

Conclusion

Ridge regression produced the lowest reported test RMSE and MAE and the highest reported R-squared. This exercise demonstrates a standard supervised-learning workflow: train/test separation, preprocessing, regularization, cross-validation, and comparison using multiple regression metrics.

Portfolio note

The accompanying R script is a cleaned, reproducible version of the original modeling exercise. The reported metrics above are preserved from the submitted analysis; reproduced values may differ slightly across package versions and resampling implementations.